

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

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Motivation

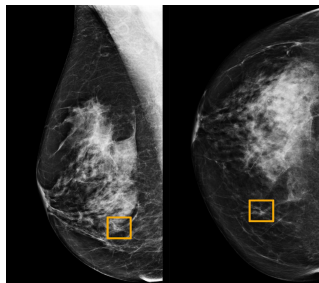
High stakes

Consider these situations:

Motivation

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Motivation

High stakes

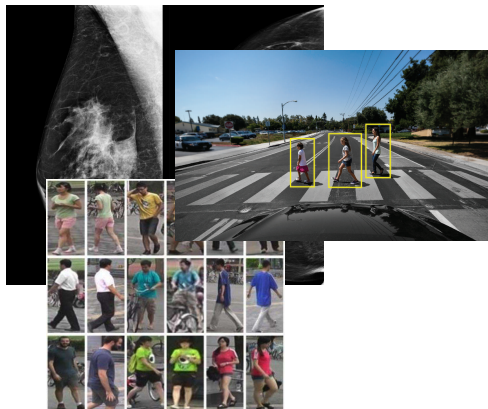
Consider these situations:



Motivation

High stakes

Consider these situations:



Straight to the point



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **understand** the behaviour of these models.

Step by step

Computation,
Computer
Vision and AI



Explainable AI



Thesis
objectives

Computation, Computer Vision and AI

Computation



Better known as *Computer Science*.

Study of:

- Algorithms.
- Data structures.
- Design of hardware and software.

Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

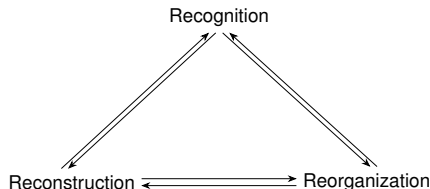
Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

Three fundamental tasks[1]:

- Recognition.
- Reconstruction.
- Reorganization.



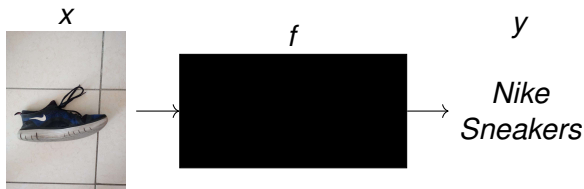
Computation, Computer Vision and AI

Artificial Intelligence

Systems capable of performing
tasks requiring human intelligence
[2].

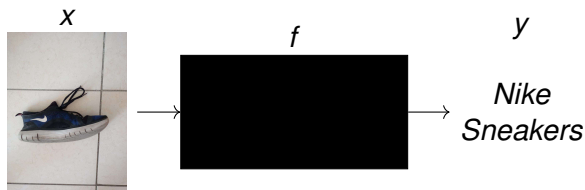
Explainable AI

We are interested in understanding models, behaving like a black box model:



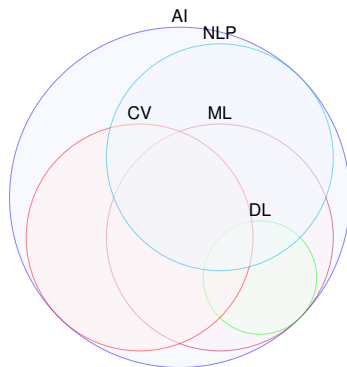
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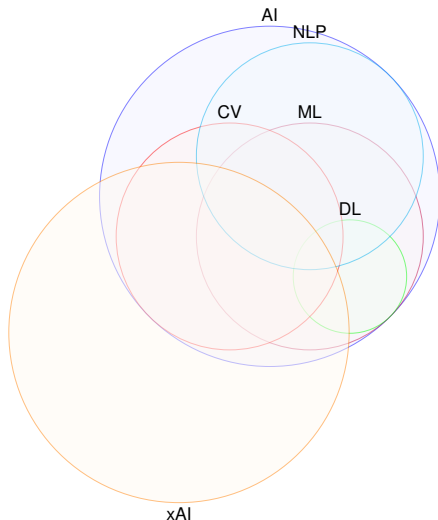


We want *know why* $f(x) \rightarrow y$

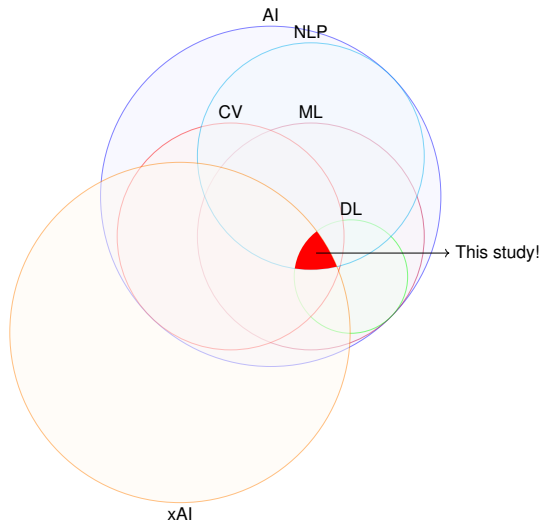
Fitting it all together



Fitting it all together



Fitting it all together



Thesis Objectives

Improvement of recognition and interpretable properties of model predictions.

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Improvement of recognition and interpretable properties of model predictions.

In particular:

- Development of low cost/complexity explainability approaches.
- Establishment of a fixed evaluation protocol.
- Differentiation of human based and machine explanations.

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Preliminaries

Desired data of Interpretability Study [4]

We ask questions
regarding *black box*
models.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?

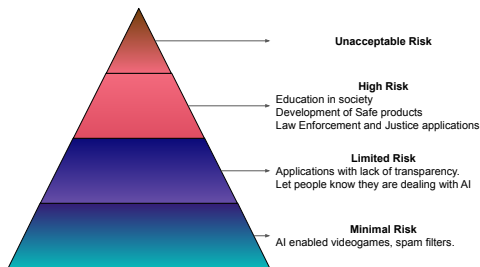
**Accepted AI systems
within society must
answer this.**

Preliminaries

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Regulation planned with the European AI act[3]

Accepted AI systems within society must answer this.

Image Recognition Models

Convolutional Neural Networks (CNN)

Based on the **convolution operation**.

A representation $f \star g$ is computed for a feature map f and a kernel g .
First approach with *Neocognitron* [5]

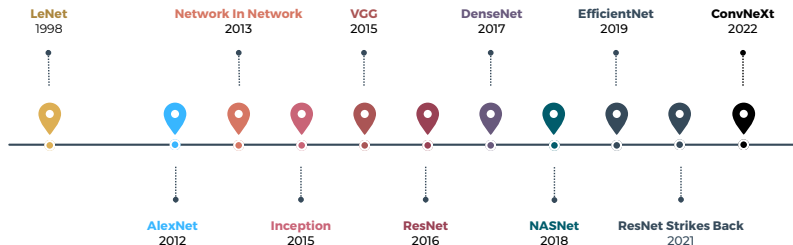
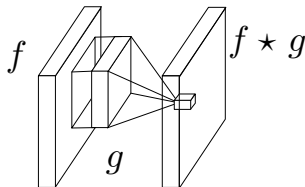


Image Recognition Models

Convolutional Neural Networks(CNN)

Training deep models is difficult, gradient information *is* lost with depth.

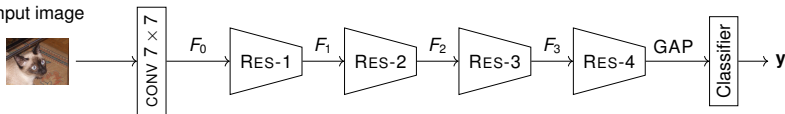
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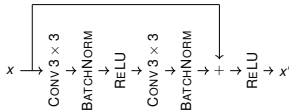
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A model surviving the test of time: *ResNet* [6].

Input image



Basic Block



Bottleneck Block

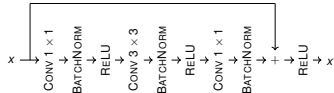


Image Recognition Models

Self Attention Architectures

Transformers

Attention: compute the relevance of each element relative to others in an embedding.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) V$$

Image Recognition Models

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Transformers

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$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) V$$

An abstract representation of a class is computed, updated with information across the embedding [7].

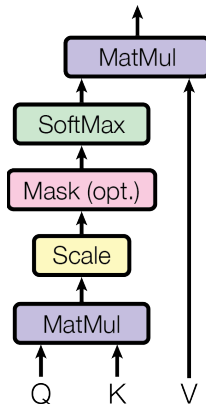


Image Recognition Models

Self Attention Architectures

A modification of transformers to vision, the *Vision Transformer* (ViT)
[8]

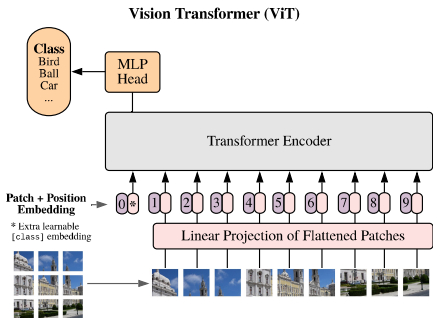


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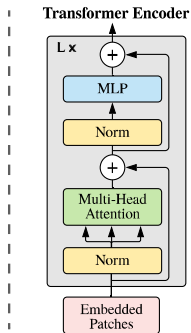
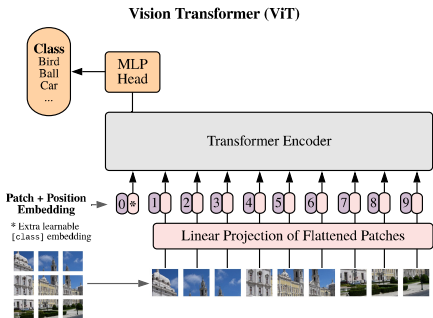
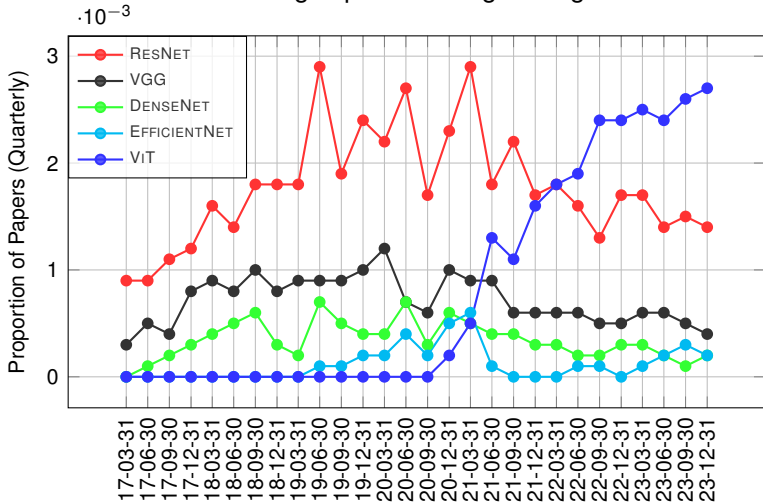


Image Recognition Models

The Current Landscape

Transformers had a strong impact on image recognition:



Interpretability

Two dimension grouping

Interpretability

Two dimension grouping

Transparency

Approaches requiring changes to an architecture or training process.

Post-Hoc Interpretability

Approaches utilizing information contained within a model.

Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



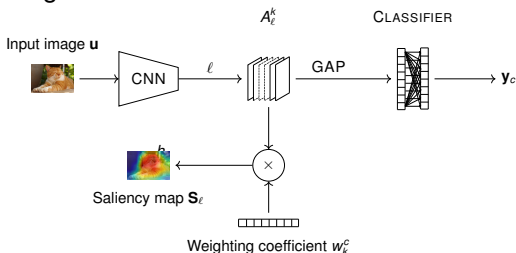
Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



Salient regions are highlighted for an input image:



Governed by equation:

$$S_\ell^c(\mathbf{u}) := h \left(\sum_k w_k^c A_\ell^k \right) \quad (1)$$

Interpretability

Evaluating Interpretability

Saliency: what is relevant towards a prediction.

An explanation should demonstrate similar predictive properties to its query.

Input image. (I)



$$P_i^c = 0.8756$$

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

Interpretability

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- Prediction loss when using an explanation. (Average Drop (AD) [10])
- Ratio of explanations being better than the input image. (Average Increase (AI) [10])
- Reaction to saliency guided modifications (Insertion/Deletion (Ins/Del) [11])

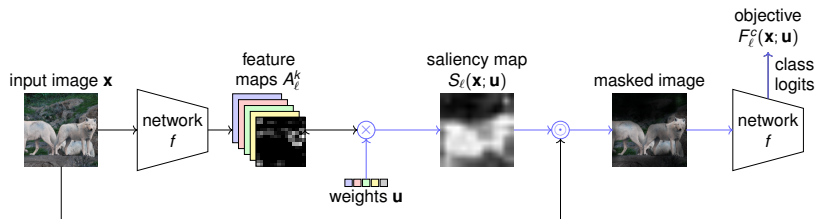
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Preliminaries

Premise

Is it possible to optimize the weighting coefficient w_k^c to obtain a saliency map with the highest probability of prediction?



Preliminaries

Contributions

Opti-CAM

Maximization of predicted probability of an explanation.

Average Drop

Evaluation metric designed to enhance comparisons.

Opti-CAM

Motivation

- **Combination of Feature Maps**

To compute w_k^c , Score-CAM [12] considers feature maps independently.

What about a linear combination?

$$F(\mathbf{w}) := f \left(\mathbf{u} \odot n \left(\text{up} \left(\sum_k w_k A_l^k \right) \right) \right)_c$$

F is optimizable with respect to $\mathbf{w}$ and an initial w_0 via backpropagation.

Opti-CAM

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F is optimizable with respect to $\mathbf{w}$ and an initial w_0 via backpropagation.

- **Masking-based Saliency** Saliency maps produced by optimization in the input space [13, 14]

The optimization usually takes form of:

$$S^c(\mathbf{u}) := \underset{\mathbf{m} \in M}{\operatorname{argmax}} f(\mathbf{u} \odot n(\text{up}(\mathbf{m})))_c + \delta R(\mathbf{m})$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (2)$$

Opti-CAM

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Find vector $\mathbf{u}^*$ maximizing confidence for class c , with an input $\mathbf{x}$ masked by Saliency Map $S_\ell(\mathbf{x}; \mathbf{u}^*)$:

$$\mathbf{u}^* := \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}^*), \quad (3)$$

The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(\mathbf{x} \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (4)$$

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The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(\mathbf{x} \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (4)$$

The Saliency Map then becomes:

$$S_\ell^c(\mathbf{x}) := S_\ell(\mathbf{x}; \mathbf{u}^*) = S_\ell(\mathbf{x}; \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u})) \quad (5)$$

Average Gain

Motivation

Existing metrics for measuring interpretability ($AD\downarrow$, $AI\uparrow$), can be fooled.

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Motivation

Existing metrics for measuring interpretability ($AD \downarrow$, $AI \uparrow$), can be fooled.

Fake-CAM [15]

What happens if you provide a Saliency Map where all the pixels *but-one* are 1?

Input



$$p_{389} = 0.874$$

Explanation



$$o_{389} = 0.822$$

● Drop = 5.95%

● Increase = 0%

Therefore, we don't drop much predictive power, *but*, we don't gain confidence in our prediction.

Average Gain

Definition

With the notation on slide (22):

$$AG(\%) := \frac{1}{N} \sum_{i=1}^N \frac{[o_i^c - p_i^c]_+}{1 - p_i^c} \cdot 100 \quad (6)$$

Measuring the increase in predictive power when masking an input.

Experimental Set-Up

Implementation

- All images are set to $224 \times 224 \times 3$.
- Adam [16] optimizer, for 100 iterations.
- Cross Entropy.
- Stop optimizing if loss is less than 10^{-10} .

¹Pytorch Model Zoo <https://pytorch.org/vision/0.8/models.html>

²Timm <https://github.com/rwightman/pytorch-image-models>

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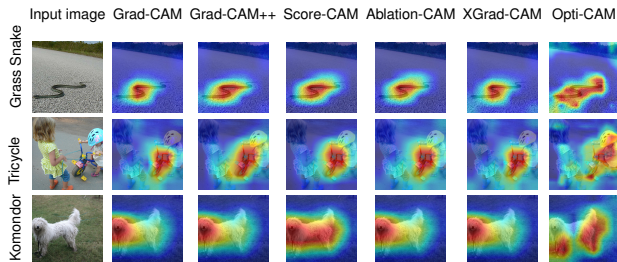
Datasets and Models

- **ILSVRC** Validation splits of ImageNet ([17, 18]), 50k images, over 1k classes.
- **Medical Data** Chest X-ray images with 3 diseases [19], endoscopy images for pathologies, findings and landmarks [20].
- **Networks** ResNet-50 [6], and VGG [21]. ¹ViT-B (16-224) [8] and DeiT-B (16-224) [22] ².

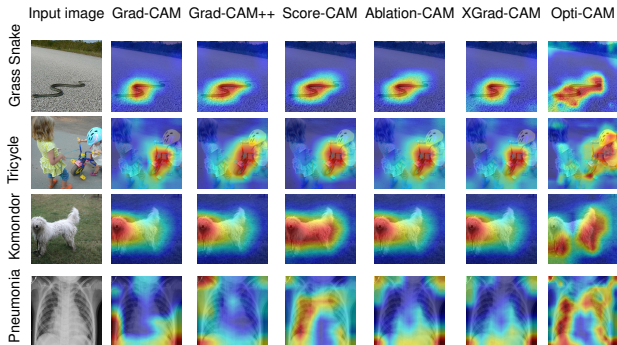
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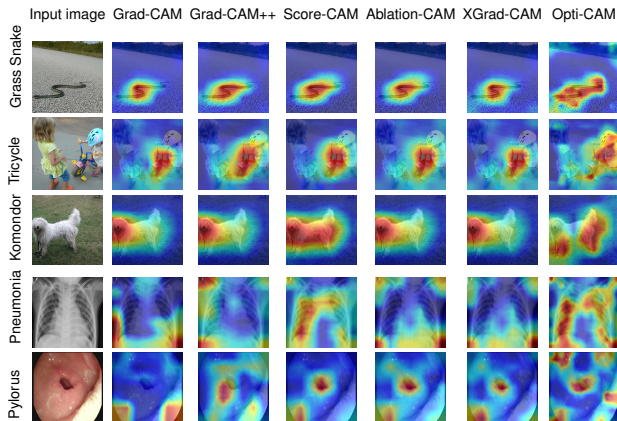
Qualitative Evaluation



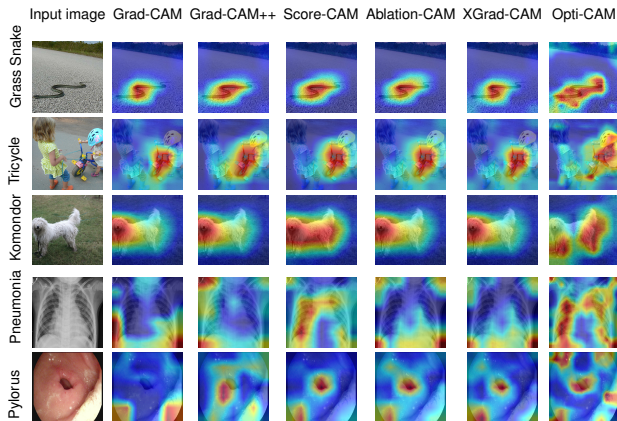
Qualitative Evaluation



Qualitative Evaluation



Qualitative Evaluation



Results

Quantitative Experiments

Classification Metrics on CNNs:

METHOD	RESNET50				VGG16			
	AD↓	AG↑	AI↑	T(s)	AD↓	AG↑	AI↑	T(s)
Fake-CAM	0.8	1.6	46.0	0.00	0.5	0.6	42.6	0.00
Grad-CAM	12.2	17.6	44.4	0.03	14.2	14.7	40.6	0.02
Grad-CAM++	12.9	16.0	42.1	0.03	17.1	10.2	33.4	0.02
Score-CAM	8.6	26.6	56.7	15.22	13.5	15.6	41.7	3.11
Ablation-CAM	12.5	16.4	42.8	18.26	15.5	12.6	36.9	2.98
XGrad-CAM	12.2	17.6	44.4	0.03	13.8	14.8	41.2	0.02
Layer-CAM	15.6	15.0	38.8	0.08	48.9	3.1	13.5	0.07
ExPerturbation	38.1	9.5	22.5	152.96	43.0	7.1	20.5	83.20

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Layer-CAM	15.6	15.0	38.8	0.08	48.9	3.1	13.5	0.07
ExPerturbation	38.1	9.5	22.5	152.96	43.0	7.1	20.5	83.20
Opti-CAM	1.5	68.8	92.8	4.15	1.3	71.2	92.7	3.94

Results

Quantitative Experiments

Classification Metrics on Transformers:

METHOD	ViT-B				DeiT-B			
	AD↓	AG↑	AI↑	T(s)	AD↓	AG↑	AI↑	T(s)
Fake-CAM	0.3	0.4	48.3	0.00	0.6	0.3	44.6	0.00
Grad-CAM	69.4	2.5	12.4	0.14	33.5	1.7	12.5	0.11
Grad-CAM	86.3	1.5	1.0	0.15	50.7	0.9	7.2	0.13
Score-CAM	32.0	6.2	33.0	23.69	53.6	2.2	12.2	22.47
XGrad-CAM	88.1	0.4	4.3	0.13	80.5	0.3	4.1	0.12
Layer-CAM	82.0	0.2	2.9	0.24	88.9	0.4	2.6	0.24
ExPerturbation	28.8	6.2	24.4	133.52	60.9	2.0	8.5	129.12
RawAtt	92.6	0.2	2.8	0.02	95.3	0.0	1.8	0.02
Rollout	42.1	5.6	20.9	0.02	55.2	0.8	7.9	0.02
TIBAV	81.7	0.8	5.8	0.16	62.3	0.7	7.1	0.16

Results

Quantitative Experiments

Classification Metrics on Transformers:

METHOD	ViT-B				DeiT-B			
	AD↓	AG↑	AI↑	T(s)	AD↓	AG↑	AI↑	T(s)
Fake-CAM	0.3	0.4	48.3	0.00	0.6	0.3	44.6	0.00
Grad-CAM	69.4	2.5	12.4	0.14	33.5	1.7	12.5	0.11
Grad-CAM	86.3	1.5	1.0	0.15	50.7	0.9	7.2	0.13
Score-CAM	32.0	6.2	33.0	23.69	53.6	2.2	12.2	22.47
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References I

-  J. Malik, P. Arbeláez, J. Carreira, K. Fragkiadaki, R. Girshick, G. Gkioxari, S. Gupta, B. Hariharan, A. Kar, and S. Tulsiani, “The three r’s of computer vision: Recognition, reconstruction and reorganization,” *Pattern Recognition Letters*, vol. 72, pp. 4–14, 2016.
-  J. McCarthy *et al.*, “What is artificial intelligence,” 2007.
-  T. Madiega, “Artificial intelligence act,” *European Parliament: European Parliamentary Research Service*, 2021.
-  Z. C. Lipton, “The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery.” *Queue*, vol. 16, no. 3, pp. 31–57, 2018.

References II

-  K. Fukushima, “Cognitron: A self-organizing multilayered neural network,” *Biological cybernetics*, vol. 20, no. 3-4, pp. 121–136, 1975.
-  K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
-  J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2018.

References III

-  A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
-  H. Lee, R. Grosse, R. Ranganath, and A. Y. Ng, “Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations,” in *Proceedings of the 26th annual international conference on machine learning*, 2009, pp. 609–616.
-  A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, “Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks,” in *WACV*, 2018.

References IV

-  P. Vitali, D. Abir, and S. Kate, “Rise: Randomized input sampling for explanation of black-box models,” *BMVC*, 2018.
-  H. Wang, M. Du, F. Yang, and Z. Zhang, “Score-cam: Improved visual explanations via score-weighted class activation mapping,” *CoRR*, vol. abs/1910.01279, 2019. [Online]. Available: <http://arxiv.org/abs/1910.01279>
-  R. C. Fong and A. Vedaldi, “Interpretable explanations of black boxes by meaningful perturbation,” in *ICCV*, 2017.
-  R. Fong, M. Patrick, and A. Vedaldi, “Understanding deep networks via extremal perturbations and smooth masks,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 2950–2958.

References V

-  S. Poppi, M. Cornia, L. Baraldi, and R. Cucchiara, “Revisiting the evaluation of class activation mapping for explainability: A novel metric and experimental analysis,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 2299–2304.
-  D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *ICLR*, 2015.
-  A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” *Advances in neural information processing systems*, vol. 25, 2012.

References VI

-  O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, “ImageNet Large Scale Visual Recognition Challenge,” *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211–252, 2015.
-  D. Kermany, K. Zhang, M. Goldbaum *et al.*, “Labeled optical coherence tomography (oct) and chest x-ray images for classification,” *Mendeley data*, vol. 2, no. 2, 2018.
-  K. Pogorelov, K. R. Randel, C. Griwodz, S. L. Eskeland, T. de Lange, D. Johansen, C. Spampinato, D.-T. Dang-Nguyen, M. Lux, P. T. Schmidt *et al.*, “Kvasir: A multi-class image dataset for computer aided gastrointestinal disease detection,” in *Multimedia Systems Conf.*, 2017.

References VII



K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” 2015.



H. Touvron, M. Cord, A. El-Nouby, P. Bojanowski, A. Joulin, G. Synnaeve, and H. Jégou, “Augmenting convolutional networks with attention-based aggregation,” *arXiv preprint arXiv:2112.13692*, 2021.